

Problem Set 1

Due on Canvas at noon Friday September 14th

Instructions: **No partial credit will be awarded.** All answers must be entered in the Problem Set 1 Quiz on Canvas. Note that it is not possible to edit your responses on Canvas after submitting, but you are able to submit a second time and your last submission is the one that will be graded. **Best practice is to complete the problem set on paper and then transfer solutions to Canvas.** You may collaborate on this problem set with other students in your learning team, but each student must submit his or her own solutions.

Grading: Grading is done on a ten point scale across all problems in the problem set. Note that after submitting on canvas, you will see a score, but that does not include any points for any qualitative questions. Grades will be finalized a few days after the due date.

Material covered: This problem set covers the first five lectures (up to Sept 11th)

Problem 1. Boycotts

Note: This problem has been a tradition at Yale SOM for many years. It is a rite of passage for SOM economics students. To answer this problem requires no mathematics beyond arithmetic. Read the instructions carefully, and then feel free to talk about this problem with your fellow students. We advise such discussions to prevent you from wasting time. In the end you must see the answer yourself in your own “aha” moment and of course, the written answer must be your own.

Setup A frequently used instrument of economic coercion is the boycott. For example, a union involved in a serious and protracted conflict with some employer or group of employers may urge all of its sympathizers to stop buying the products of that employer or group, in the hope that the resulting economic losses will provide an incentive for the employers to make concessions. However, experience shows that the coercive power of a boycott is often small.

In this exercise, you are to use supply and demand analysis as a tool to understand the effects of boycotts. For the sake of concreteness and precision, most of the questions are posed in quantitative terms. However, the goal of the exercise is to develop your understanding of the logic of the market processes involved. Accordingly, if you have difficulty, don't try to overwhelm the problem with calculation -- rethink the economics.

Throughout the exercise, the following assumptions are to be maintained. First, the market in question is one to which simple supply and demand analysis is applicable: there are numerous buyers and sellers in the market; the good is homogeneous; buyers and sellers seek only to trade on the most advantageous possible terms and, aside from the complications introduced by the boycott, do not care with whom they transact. Second, the identities of buyers and sellers are fixed: in particular, no reselling of the goods takes place. Third, the set of buyers divides sharply into observers and nonobservers of the boycott, while the sellers separate equally sharply into those that are targets of the boycott and those that are not. Boycott observers will not, under any circumstances, buy from a target seller, while non-observers do not care from whom they buy. For simplicity, it is assumed that boycott observers account for the same fraction of demand at every price. Similarly, target sellers account for a certain fraction of the supply at every price.

Assume initially that in the pre-boycott situation, the market demand curve is described by:

$$Q_D = 5000 - 10P$$

where Q_D = quantity demanded and P = the price. The market supply curve has equation (with Q_S = quantity supplied):

$$Q_S = 50(P - 5)$$

except that, at prices of 5 or less, quantity supplied is zero.

Warmup problem (i): Determine the market equilibrium in the pre-boycott situation.

Answer: In equilibrium, the price is such that quantity demanded equals quantity supplied:

$$\begin{aligned}
Q_D &= Q_S \\
5000 - 10P &= 50(P - 5) \\
60P &= 5250 \\
P &= 87.5 \\
Q &= 4125
\end{aligned}$$

Warmup problem (ii): Suppose that buyers representing 70% of the original demand, observe the boycott. If all non-observers buy from target sellers, what is the demand curve facing target sellers?

Answer: Since observers will not buy from target sellers at any price, the only demand facing target sellers is from non-observers. Since observers and non-observers account for demand in the ratio 0.7:0.3 at every price, the equation of the demand curve facing target sellers is:

$$Q_D = 0.3(5000 - 10P) = 1500 - 3P$$

Questions

Consider the three situations characterized by the following table

<u>Situation</u>	<u>Observers as Fraction of Original Demand</u>	<u>Target as Fraction of Original Supply</u>
A	3/4	3/4
B	1/4	1/4
C	1/2	1/4

For each of the three situations, determine the market equilibrium that would arise under boycott conditions. Give the quantities transacted by observing and non-observing buyers, and by target and non-target sellers.

Drawing on your results, state the general principle which relates the effectiveness of a boycott to the observance fraction and the target fraction. "Effectiveness," in this context, means lowering the price to target sellers. Can you provide some intuition as well?

Hint: Warmup problem (i) supplies all the mathematical knowledge you will need to solve the remaining problems. What is needed for the rest is an ability to structure the problem, and an understanding of simple economic principles. The point of the exercise is to determine whether the boycott is effective in any of the three cases above. To answer this question, you have been given the following information: (1) the market supply curve, (2) the targeted fraction of sellers, (3) the market demand curve, (4) the observers as a fraction of buyers. Items (1) and (2) imply that it is possible to write down algebraic expressions for the supply curves of targeted and non-targeted sellers. You can also choose to use Excel to get a sense of how prices get determined in each sub-market.

Problem 2. A Problem of Ice and Fire

- a. In the land of Westeros, the main source of iron ore is the “Iron Islands”, where ore is supplied from mines according to the following annual supply function:

$$Q_S = \frac{1}{3}(1,000P - 40,000)$$

Annual demand comes from all types of industries across Westeros and is given by

$$Q_D = 70,000 - 500P$$

Solve for the equilibrium price of a ton of ore in Westeros and the annual quantity produced.

Price: _____

Quantity: _____

Answer. Equating supply and demand yields a price of \$100 and quantity of 20,000 tons.

- b. Cersei, the Queen of Westeros, has recently had a falling-out with House Tyrell, who had been a major financial supporter. As a result, she is considering imposing a \$10/ton tax on iron ore sales to raise revenues. How much government revenue would be raised by such a tax?

Annual tax revenue: _____

Answer. Tax revenue is \$10 times quantity, or \$180,000. Equilibrium quantity comes from equating the tax-adjusted supply curve with demand. Tax-adjusted supply comes from first solving for inverse supply and then adding \$10:

$$P_S = 40 - \frac{3}{1000}Q + 10$$

- c. Far to the east lies the city of Braavos and the much larger continent of Essos, where the prevailing price of iron ore is \$70. The entrepreneurial merchants of Braavos decide to establish a trade route with Westeros, which leads to global free trade in iron ore. Assume trade is costless, Westeros is “small” relative to Essos, and that Queen Cersei has not imposed the tax she considered above. How much ore will be imported to or exported from Westeros? (Without doing any more math:) Has trade made iron consumers in Westeros better off or worse off? What about producers?

Quantity (circle one) IMPORTED EXPORTED to/from Westeros: _____

☐ Consumers and producers better off

☐ Consumers better, producers worse off

☐ Consumers worse, producers better off

☐ Consumers and producers worse off

Answer. At a world price of \$70/ton, domestic demand is 35,000 tons, while domestic supply is 10,000 tons. Demand is greater than, and price is lower than, autarky levels. Therefore, 25,000 tons will be imported to Westeros. Consumers are better off and producers are worse off as prices have effectively fallen to \$70.

- d. The producers in the Iron Islands have a proposal for Queen Cersei: given trade levels, why not impose an import tariff of only \$9/ton on any iron ore imported from Braavos to raise the revenues she needs? Solve for the amount of tax that would be collected from such a tariff so that the Queen can evaluate this proposal relative to imposing a tax and barring all trade.

Annual tariff revenue: _____

Queen Cersei, who cares only about government revenue, should implement:

☐ Import tariff of \$9

☐ \$10 Tax and trade ban

☐ Unclear

Answer. The tariff would effectively raise the world price to \$79, where imports would now be 17,500 tons, and so tariff revenue would be \$157,500. This is far less than Cersei needs, and she should never listen to these producers or their lobbyists.

- e. (Without doing any more math:) what would be the change in producer surplus for Westeros' Iron Islands (relative to free trade), the change in consumer surplus for the customers in Westeros (relative to free trade) and the effect on total surplus for Westeros if this import tariff were introduced?

Westeros producer surplus (under the tariff vs free trade) would ...

☐ Increase

☐ Decrease

☐ Unclear without more information

Westeros consumer surplus (under the tariff vs free trade) would ...

☐ Increase

☐ Decrease

☐ Unclear without more information

Westeros total surplus (under the tariff vs free trade) would ...

☐ Increase

☐ Decrease

☐ Unclear without more information

Why did you conclude this about total surplus? _____

Answer. Producers would be better off under the tariff, consumers would be worse off, and total surplus would unambiguously decrease as the tariff would introduce deadweight loss.